

REPORT
OF THE
STATE VETERINARIAN
OF
CALIFORNIA

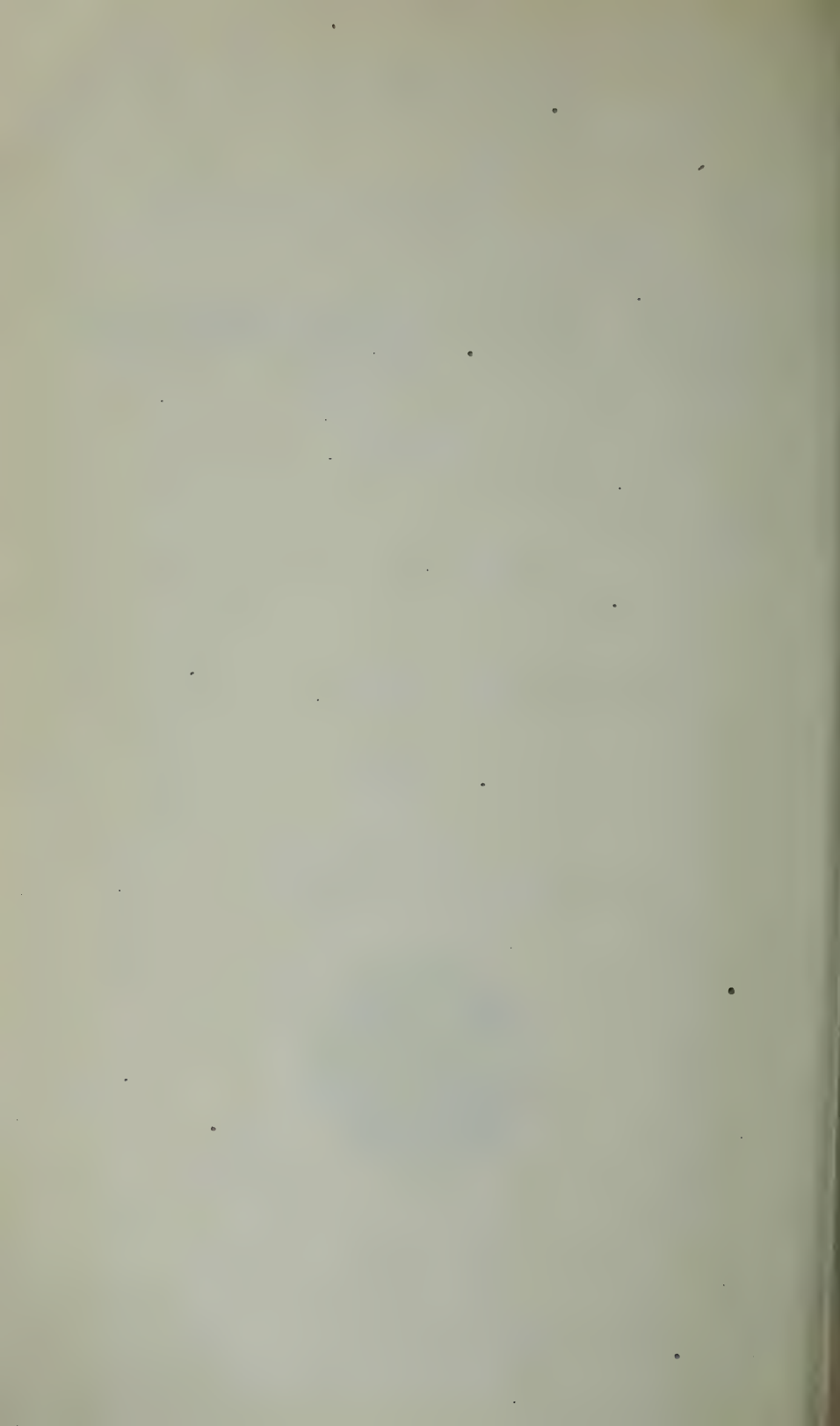
For the Two Years Ending June 30, 1908.

CHARLES KEANE, State Veterinarian.



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REPORT OF THE STATE VETERINARIAN.

*To His Excellency, J. N. GILLETT,
Governor of California.*

DEAR SIR: I respectfully beg leave to submit my report as State Veterinarian for the fifty-eighth and fifty-ninth fiscal years. Unfortunately, on account of the inadequacy of our office force, we are unable to present accurate statistics of the work that has been accomplished through the efforts of this department for the past two years. However, in several lines the results have been very gratifying. I refer especially to cattle tick and sheep scab eradication.

New avenues, whereby this department can be of material assistance to the livestock interests, are being opened up, but this new work must necessarily be, more or less, neglected on account of work on hand. For example, glanders, a most virulent infective disease of equines, and also communicable to man, exists in many counties in this State. Wherever our attention is called to the existence of this disease in any locality, measures are adopted whereby the affected animals are destroyed and surrounding premises cleaned and disinfected. However, this disease is constantly increasing among our horse and mule stock, and this can only be prevented by a systematic course of inspection in the counties in which the disease is known to exist.

The same might be said of the disease known as anthrax. This disease is the cause of extensive fatalities on some of our cattle ranches situated in the Sacramento and San Joaquin valleys. In order to check the further spread of these animal plagues a complete system of veterinary sanitary police must be adopted.

Furthermore, during the early part of the present century several of the New England states were visited by a serious animal plague. I refer to contagious foot and mouth disease, which affects cattle, sheep, hogs, and other animals. Before this disease was finally exterminated the livestock interests in these states suffered serious financial losses. This disease was introduced from Europe, either by means of diseased stock or through intermediate bearers of infection. (At the present writing, November, 1908, foot and mouth disease has again been introduced into New York, Delaware, Maryland, and Pennsylvania from the outside.) California is so situated that it is possible that disease in the shape of animal plagues might at some time be introduced from

the Orient, where such diseases devastate livestock. For these reasons it is essential that this department be organized in a manner to combat and exterminate our prevalent diseases and also to prevent, as far as possible, the introduction of exotic diseases.

The day of the large stockowner in California is nearing an end. Already many of the large ranches in the valleys, where for years great herds of cattle, sheep, and horses were kept, are being cut up into smaller tracts. In years past the large ranch owner looked upon the loss of a certain percentage of his livestock as a natural event, and his possessions were so extensive that he did not feel the loss. Not so with the small ranch owner with his fewer number of livestock. He cannot afford to lose any, and a serious loss may cripple him financially. He needs protection and advice, and it is for this purpose that this department was created.

With the cutting up of the extensive land grants into smaller parcels, the work of this department increases. This is well exemplified in the fight to exterminate the cattle tick. For example, the Laguna de Tache land grant, located in the western part of Fresno County, along the Kings River, is tick infested land. For several years past small parcels of this grant have been sold to settlers, the majority of whom conduct dairies. Their cows became infested with the cattle tick, and, instead of the one large infested ranch, we have numbers of small ones. On the large ranches the work of eradicating the ticks is easier, inasmuch as the starvation method can be resorted to; while on the smaller infested ranches the tick must be fought by continued disinfection of the cattle. In order to see that this work is properly done a continual official supervision of the work becomes necessary.

SHEEP SCAB.

In March, 1908, the work of eradicating scabies in sheep was inaugurated by this department. Although improperly equipped for the successful carrying out of a work of this magnitude, it was seen that something should be done to eradicate and check the further progress of this disease. The United States Department of Agriculture, through the Bureau of Animal Industry, was called upon for assistance in the shape of inspectors to coöperate with this department in the work of supervising the dippings of sheep. This assistance was promised to us with the provision that the State of California would do its share of supplying a certain number of inspectors. As we had no appropriation available for such work, the task of requesting the various boards of supervisors, in counties in which sheep are located, to make suitable appropriations in order to pay for the services of State inspectors, was resorted to. On the whole, the various interested counties

acted promptly, so that the State was enabled, during the period from March 15th to June 30th, 1908, to place at different times an average of 65 inspectors in the field. The aggregate amount of money appropriated by the different counties was approximately \$25,000. While the majority of these men made excellent inspectors, a number of them were inexperienced in this work; consequently, it was some time before they were competent to properly carry on the work in the respective districts to which they were assigned. In the same period of time the Federal Government placed in the field 40 inspectors, many of whom were experienced men in this line of work. The Federal inspectors worked under the supervision of a local inspector in charge.

Before starting the work the Federal inspector in charge and the state veterinarian adopted plans whereby the work would be conducted systematically and thoroughly. It was decided to divide the state into districts and place in charge of each district an experienced federal inspector, who supervised the work of both federal and state inspectors in his respective district. A system of inspectors' reports was issued to the inspectors, and they were required to send these reports to Sacramento on the completion of each day's work. These reports showed in detail the number, condition, location, and owner of the sheep dipped, as well as the kind of parasiticide used in the dipping.

On March 2, 1908, Your Excellency, at the request of the State Veterinarian, issued a proclamation to the effect that between the dates of March 15, 1908, and July 15, 1908, all sheep in the State of California would be required to be dipped by the State Veterinarian or his duly authorized agent, in solutions sanctioned by said officials. It was decided to dip unexposed, clean sheep only once, and sheep exposed to, or infected with scabies, twice. This general dipping of all the sheep in the State was considered necessary in the inauguration of this work on account of the fact that preliminary inspections of flocks in different localities showed that a large proportion of our sheep were infected with scabies. From a compilation of inspection and dipping reports it was shown that about 65 per cent of the entire number of sheep in this state were infected. In some instances this infection was mild, but in many cases the disease was in an advanced stage.

Ideally it would appear that a general dipping, such as our sheep were subjected to, would completely eradicate scab. But in a work of this magnitude it is impossible to get ideal results. However, scabies has been greatly reduced; so much so that the further dipping of clean sheep is not considered necessary. Attention will be paid to infected and directly exposed sheep.

A number of obstacles were met with in the conduct of the work. Feeding conditions were bad. California did not receive her normal

quota of rainfall. In the San Joaquin and Sacramento valleys practically no late rains fell. Consequently the feed was short and ripened early. The calls for inspectors to supervise dippings in conformity with the proclamation seemed to come in all together. Sheepmen wished to comply with the regulation and then move sheep to the summer ranges early, in order to get new feeding grounds and to prevent damage to lambs from foxtail. In other places both feed and water were short. Eventually enough inspectors were appointed to attend to all official work.

Following is a summary of the sheep dipped between March 15 and June 30, 1908, showing number of infected and free sheep, as well as the number of sheep dipped once and twice, and the dippings supervised by State and Federal inspectors.

Number of infected sheep.....	2,278,860	
Number of free sheep.....	702,761	<u>2,981,621</u>
Number of sheep dipped twice.....	2,243,354	
Number of sheep dipped once.....	738,267	<u>2,981,621</u>
Number of sheep dipped by State inspectors once.....	260,605	
Number of sheep dipped by State inspectors twice.....	1,328,900	1,589,505
Number of sheep dipped by Federal inspectors once.....	477,662	
Number of sheep dipped by Federal inspectors twice.....	914,454	<u>1,392,116</u>
		<u>2,981,621</u>
Total dippings by State inspectors.....	2,918,405	
Total dippings by Federal inspectors.....	2,306,570	
Total dippings for four months ending June 30, 1908.....		5,224,975

NOTE.—The discrepancy in the figures between number of infected sheep (2,278,860) and number of sheep dipped twice (2,243,354) is due principally to the fact that after one dipping some of these sheep were sold for slaughter.

While many of the interested counties coöperated with the State in sheep scab eradication work in the spring of 1908, it is very improbable that we will receive the same coöperation throughout 1909. Furthermore, it would be very desirable for the state to have at its command a sufficient number of temporary inspectors in order that this work may be completed as soon as possible. For this reason I would respectfully recommend that an emergency appropriation be granted this department in order to employ ten inspectors for the period between March 1st and June 30th, inclusive, of this fiscal year.

I would also recommend an appropriation of \$12,000 to pay for the service and traveling expenses of ten temporary sheep inspectors to serve whenever required during the sixty-first fiscal year, at the end of which time I believe this State will be practically free from sheep scabies.

CATTLE TICK ERADICATION.

The chief efforts of this department for the past two years have been directed to a continuation of the fight against the southern cattle fever tick (*Boophilus annulatus*). Considering the nature of the work and the antagonism we formerly met with in some sections in the conduct of this fight, the results have been very gratifying. Where we formerly met with considerable opposition, this has now been reduced to a minimum. Several years ago the work of eradicating the cattle tick was considered a hopeless task, even by many of the most intelligent cattlemen. Now, with practically few exceptions, we have the hearty coöperation of the cattlemen in this work and as a result rapid strides have been made in this line.

In his annual report to the secretary of the United States Department of Agriculture for the year 1907, the chief of the Bureau of Animal Industry says, "Such good results were obtained in California (in the eradication of the cattle tick) that an area of about 20,000 square miles was released from quarantine August 1, 1907, and it is believed to be only a question of a short time when the infection in California will be confined to a comparatively small area along the southern coast."

The area of quarantine territory in the United States due to cattle tick infestation, stretches clear across the southern part of the country from California to Florida, omitting Arizona and New Mexico. In all this territory, from January 1 to October 31, 1907, 40,970 square miles of territory were cleaned and removed from quarantine, of which 21,486 square miles were in California.

The importance of exterminating the cattle tick cannot be denied. It is estimated that the losses in the United States, directly and indirectly, due to the cattle tick, amount to more than \$25,000,000 annually. In the past, California has shared her quota of this loss. At the present time, however, the loss in California is reduced to a minimum. This is due to the fact that rigid quarantines are maintained on every infested ranch and all infested ranches are in process of cleaning. Furthermore, during the two years last past there has been no extension of the infestation to points outside of the quarantined area, due to maintenance of our state quarantine, which prevents the movements of all cattle from the quarantined area to points outside of same.

In response to our plea the legislature of 1907 wisely enacted a law giving the State Veterinarian authority to seize and disinfect cattle for the purpose of eradicating ticks therefrom in cases where owners refused to follow our instructions. The conferring of this authority upon the State Veterinarian became necessary from the fact that a few recalcitrant owners of tick infested cattle refused to adopt measures

to eradicate the ticks on same. Their cattle being immune to Texas fever and not being required to move their stock for feed, quarantines had no effect on them. At the same time surrounding clean ranches were constantly being exposed to infestation. It became necessary to apply this law in several instances, two of which occurred in San Luis Obispo County. In suing one party to collect expenses the state received a decision in the Superior Court of San Luis Obispo County. In the other case the lien was collected without the matter going to trial.

On July 15, 1906, by proclamation issued by the Governor, all of Merced County, including two large ranches that extend over into Madera County, was removed from quarantine.

On February 1, 1907, there was removed from quarantine all that area of territory in Madera and Fresno counties situated east of the right of way of the main line of the Southern Pacific railroad. At the same time Santa Barbara County was placed in the provisionally quarantined area, which allows the movements of cattle from Santa Barbara County at all times of the year after they have been inspected and found free from cattle ticks. The work had progressed so well in this county that the movement of cattle after inspection was considered safe.

On August 1, 1907, there was removed from quarantine Kings, Tulare, Kern, and that portion of San Bernardino County situated east of the one hundred and sixteenth meridian west longitude. On the same date Los Angeles County and that part of San Bernardino County situated west of the one hundred and sixteenth meridian west longitude, were placed in the provisionally quarantined area, from which cattle may be removed at all times of the year after they have been inspected and found free from cattle ticks.

On February 1, 1908, the work of cattle tick eradication had progressed so well in Ventura and Riverside counties that they were also placed in the provisionally quarantined area, from which cattle may be moved at all times of the year after they have been inspected and found free from ticks.

To sum up: San Diego, Imperial and Orange counties are at present the only strictly quarantined counties left in the state.

San Luis Obispo, Santa Barbara, Ventura, Riverside, and Los Angeles counties, that part of Fresno County situated west of the right of way of the Southern Pacific railroad, and that part of San Bernardino County situated west of the one hundred and sixteenth meridian west longitude, are in the provisionally quarantined area. One, and possibly more of these provisionally quarantined counties is now also clean and will shortly be entirely removed from quarantine restrictions.

The work has also progressed so well in San Diego, Imperial, and Orange counties that they will shortly be placed in the provisionally quarantined area.

Two specific appropriations have been made by Congress to the United States Department of Agriculture in order to enable that department to coöperate with the authorities in the different tick infested states in the work of cattle tick eradication. Without this assistance California would not have made the progress she has. The inspectors of the Bureau of Animal Industry, acting under the supervision of a chief inspector, Dr. W. M. MacKellar, working with the county and state officials, devoted their entire time to this work, uncovering infestation and supervising dippings in accordance with state laws.

A number of methods for the eradication of the ticks have been used. These included pasture rotation, dipping, spraying, and hand dressing. It has been our practice to recommend the least expensive and most advantageous method to suit each individual case. For instance, on some of the large ranches, where large herds are run, the starvation method is usually adopted. This is the most desirable method and is adopted from our knowledge of the life history of the tick. Cattle, horses, and mules are taken off of any part of an infested pasture, which is then fenced off and allowed to remain idle for a certain length of time, varying, to the time of year the cattle were taken off, usually about eight to ten months. In this time all the ticks that were left on the pasture, after removal of the cattle, will die. The cattle are then thoroughly dipped in a parasiticide in order to kill all ticks on them and are then placed on the pasture that has been idle. Then the stock is removed from another field in a similar manner.

At times an entire ranch is rested up and cleaned in one year. At other times one half, one third, or as much of the ranch as can conveniently be sacrificed at one time is cleaned.

Sheep are also utilized in the work of cleaning large ranches. While in a few rare instances several cattle ticks have been found on sheep, for all practical purposes we have found that these animals can be placed on an infested field after cattle, horses and mules have been taken off, and after a period of from eight to ten months all ticks on the fields will have died. At the expiration of this time clean cattle can be placed on these fields with comparative safety. In this manner no feed is sacrificed. The sebacious secretions in the skin of sheep appear to be obnoxious to the cattle tick. On the small infested ranches, where pasture rotation is out of the question, dipping in a parasiticide solution is resorted to. This method, while slower than pasture rotation, is also adopted from our knowledge of the life history of the tick. The aim is to prevent regenerations of ticks, and this

is accomplished by dipping all cattle on an infested ranch every three or four weeks during the summer, fall, and early winter months. The tick infests cattle as a small, minute seed tick. It sucks the blood of the animal and, after undergoing two moults, the female enlarges until it reaches maturity at the end of three to six weeks, at which time she loosens her hold and drops to the ground and lays several thousand eggs. After a certain period, according to the weather, the majority of these eggs hatch and the young seed ticks emerge and again infest cattle. In order to prevent regeneration of ticks the cattle are dipped every three or four weeks in order to destroy female ticks nearing maturity on the animals, thereby preventing further reproduction.

It was a difficult matter for a long time to find a suitable parasiticide that would kill the ticks without injuring the cattle. Prior to 1907 crude petroleum oil floated on water in a dipping vat was used. Infested cattle were made to swim through this liquid. While the oil was very effective in destroying the ticks, it was also found that in many cases it would injure cattle by blistering the skin. This was especially evident in the case of milk cows in which the skin of the udder was often blistered, resulting in a falling off, and at times a suppression, of the milk yield.

The arsenical dip was first used in California during 1907, and such good results were obtained that it has been our most effective remedy ever since. The formula used is:

Arsenic	8 pounds
Carbonate of soda	24 pounds
Soap	24 pounds
Pine tar	1 gallon
Water	500 gallons

This solution, being poisonous in nature, is carefully used in order that cattle dipped in same are not poisoned themselves. However, I might state that no authentic case of cattle poisoning from this dipping solution has been brought to our notice. In several instances it was reported that cattle dipped in the arsenical solution had been poisoned, but upon investigation it was found that deaths were due to other causes. One notable case occurred in Orange County, where a number of cattle were dipped in this solution preparatory to moving them on to a tick free pasture. Subsequently several head of these cattle died from what was claimed to be arsenical poisoning, due to the cattle drinking the dip while swimming through the vat. Upon autopsy not the slightest indication of arsenical poisoning could be found. The writer, in company with the Assistant State Veterinarian, personally investigated this matter, and it was ascertained that these cattle, the day they were dipped, were driven a distance of some 30 to 35 miles. The cattle were worked too hard in one day, and most probably died from exhaustion.

In another instance several cows died on a ranch in San Luis Obispo County after dipping. Autopsy by a Federal inspector revealed in these cases well marked lesions of Texas fever, with no indications of arsenical poisoning.

Reports were received at this office during the past two fiscal years of tick infested cattle located in Monterey and San Benito counties, which territory is in the free area. Upon inspecting the cattle upon these ranches we found them to be infested with the common net tick (*Dermacentor reticulatus*), and to be entirely free of the southern cattle fever tick (*Boophilus annulatus*).

During the last fiscal year two prosecutions were instituted by the state for infraction of the state quarantine regulations. In both cases convictions were secured.

EPIZOOTIC CEREBRO-SPINAL MENINGITIS (SO-CALLED).

This affection was the cause of considerable loss among horses and mules in widely separated parts of the State during the past two years. Outbreaks were investigated by the State and Assistant State Veterinarian in Siskiyou, Contra Costa, Colusa, Merced, and San Diego counties. The disease was confined to one or two ranches in each of the counties mentioned.

From the suddenness of the outbreak and the number of animals affected this affection assumes the characteristics of a communicable disease, and in fact some authors consider it similar to the infectious cerebro-spinal meningitis of man. Fortunately the affection in horses and mules is not infectious in character, but is due to food that has become infested with poisonous moulds, and also at times possibly to contaminated water. The disease can easily be produced experimentally. In every instance we have found no difficulty in checking the further spread of this affection by proper hygienic measures.

GLANDERS.

The existence of this disease among horses and mules in various sections in California is a serious menace to our horse and mule industry as well as to public health. Several deaths in man, due to infection communicated by glandered horses, have been reported during the past two years. From time to time this department has investigated and destroyed horses and mules affected with glanders in a number of counties in the state. Unfortunately, no records were kept. However, we can safely say that the disease exists in many sections of the state. While every reported outbreak of this disease is investigated, and the diseased animals killed and the premises disinfected, in conformity with section 402b of the Penal Code, yet this procedure is only palliative, as only a small percentage of cases are reported to this office.

This disease has an economic importance to the stockmen, and it has also a sanitary aspect to the general public, as it is readily communicable to man and is often fatal.

The disease is so widely scattered in the State that only drastic and thoroughly systematic measures will check its further spread and exterminate it. Unfortunately, in many instances, owners of animals affected with the chronic form of this disease will hide its existence by various means. In many instances, also, the affected animals are able to work and are constantly spreading infection. It is only by systematic effort that this disease can be exterminated. This means a thorough ranch to ranch and stable inspection of all horse and mule stock in the state. To do this at present is out of the question, due to lack of facilities. This department should be provided with sufficient help and funds to carry out this work to a finish.

Glanders is similar to tuberculosis in cattle in one respect, inasmuch as an animal may be infected with this disease and exhibit no external manifestations. It is a popular fallacy that an animal affected with glanders should exhibit external symptoms, such as a typical nasal discharge, swelling of the sub-maxillary lymphatic glands, etc. On the contrary, we find many cases in which the lesions are located in the internal organs and exhibiting no characteristic external lesions. These are dangerous factors in the spread of the disease. Fortunately we have a means of diagnosing glanders where we have no external symptoms to guide us. The mallein test properly conducted by a competent operator is a reliable and accurate means of diagnosing occult cases of glanders in horses and mules.

While section 402b of the Penal Code makes it a misdemeanor for a person to have in his possession an animal affected with glanders or farcy, I believe this law should be amended so as to give explicit authority to the State Veterinarian to destroy animals affected with this disease. Legislation should also be enacted requiring veterinarians to report all cases of glanders to the county health officers, who in turn would be required to notify the State Veterinarian.

ANTHRAX.

This disease was the cause of serious losses to our livestock in a number of sections of the state during the past two years, as it has been for years past. It is practically impossible to eradicate this disease from an infected ranch by ordinary sanitary measures. The spore or seed of the organism that is the cause of this disease is the most resistant disease-producing organism with which the scientific world is familiar. As far as we know it will live for years in the soil, and for all practical purposes a ranch once infected will remain so indefinitely. On a number of California ranches anthrax appears regu-

larly every year and is the cause of serious losses. The owners of some of the larger ranches are able to move their cattle from the infected fields early in the fall and bring them back late in the winter, and thereby prevent serious loss. The disease usually appears when the feed is short, probably due to the fact that infection exists close to the grass roots. However, on a number of the smaller ranches the livestock is allowed to run on the fields during the entire year, and on these places preventive vaccination is resorted to. Preventive vaccination against anthrax is not always successful. However, in many instances it reduces mortality.

In company with the livestock inspector of San Bernardino County the State and Assistant State Veterinarian investigated a serious outbreak of this disease on a large dairy ranch near the city of Redlands. The fact that anthrax can be communicated to man, and is often fatal, makes the existence of this disease on a dairy farm a very serious matter. Fortunately, the livestock inspector of San Bernardino County had been advised of the presence of this disease early enough to prevent the possibility of the extension of the same to consumers of milk by prohibiting its sale.

From the fact that anthrax usually makes its appearance during the fall months, reports of several outbreaks often reach this office at the same time. For this reason it is physically impossible for us to devote sufficient time to each outbreak. Very often it is necessary to handle an outbreak by correspondence. This is, of course, very unsatisfactory. Our instructions regarding the cremation of carcasses are not always followed out for the reason that there is no official on the ground to supervise the work. Where a carcass is not properly disposed of there is always the great danger of the extension of the infection to neighboring ranches.

SYMPTOMATIC ANTHRAX.

This disease, also called black leg, black quarter, etc., was in former years the cause of severe losses. Preventive vaccination has robbed black leg of its terrors, and is almost universally practiced by stockmen in this state. Where reliable vaccine is used and the operation conducted properly fatalities are reduced to a minimum.

A number of outbreaks of this disease among young cattle were reported to this department during the past two years, and which were immediately checked by following our instructions.

HOG CHOLERA.

Reports of this disease were received from Shasta, Tehama, Solano, Tulare, Kings, and Los Angeles counties, in each of which the losses were large. This disease is very infectious in character and the mor-

tality is very high. Up to a short time ago it could only be combatted by the most rigid sanitary measures, such as separation of the healthy from the sick, cremation of carcasses, cleansing and disinfection, in conjunction with quarantines. The United States Department of Agriculture has made an exhaustive study of this disease with a view to determining the specific cause and, if possible, discover a preventive agent. That these experiments appear to have been successful to a certain degree has been shown in a report issued by the department in which the results of the experiments show quite clearly that a comparatively certain method of protecting hogs from this disease has been secured.

It has been proven that the contagion of hog cholera exists in the blood and other fluids of diseased animals, but this contagion is invisible under the microscope, and can also pass through the finest filter.

Hogs that have recovered from cholera (acquired immunes) and hogs that have lived through an outbreak without contracting the disease (natural immunes) are injected with suitable amounts of virulent blood from diseased hogs and their immunity is thus heightened (hyperimmunization). The blood serum of the immunes is then injected under the skin of the hogs which it is desired to protect. The method giving the best results is to inject simultaneously blood from diseased hogs with the immune serum.

In order to demonstrate the above method, the Chief of the Bureau of Animal Industry invited the officials in charge of animal sanitation in a number of states to a demonstration of this process in order that the various interested states might take up this work. The State Veterinarian of California, who was invited, was compelled to decline the invitation, on account of important work on hand in the state at that time. However, it is essential that this method of immunizing hogs against cholera be introduced into California at the earliest possible date.

TUBERCULOSIS.

During the past two years there has been considerable agitation in California with reference to the improvement of its milk supply. Reports to this office from the inspectors of the United States Bureau of Animal Industry, in charge of meat inspection at our large abattoirs, of condemnations of cattle for tuberculosis, as well as results of tuberculin tests made through this office and private veterinarians, lead us to conclude that this disease exists more or less generally among our dairy herds. On some of our beef ranges the disease also exists among cattle. It has long been presumed that the outdoor life of range cattle precluded the possibility of their contracting tuberculosis, but abattoir inspection has revealed the presence of this disease in a number of our range herds.

The percentage of tuberculosis in California hogs is also increasing alarmingly. Hogs appear to be particularly susceptible to this disease. It is not uncommon for a considerable number of hogs in one shipment to be condemned in the abattoir for tuberculosis. That hogs contract the disease from cattle is indisputable. It has been demonstrated that the commonest mode for the discharge of tubercle bacilli from the bodies of cows is with their feces. It has also been demonstrated that the sediment taken from cream separators often contains tubercle bacilli. Of course, in these cases the milk came from tuberculous cows. Hogs are fed on this separator milk, and if it is contaminated with tubercle bacilli they readily contract the disease.

While it is universally conceded that tuberculosis of man and cows are identical, differing only in certain morphological characteristics of the tubercle bacilli, and that it can be transmitted to humans through infected milk of cows, this disease has also an economic importance to the stock raiser. Ward and Haring, of the University of California, in their instructive and interesting bulletin entitled "Bovine Tuberculosis," say:

"The importance of bovine tuberculosis, as an economic problem in agriculture, has too frequently been clouded by a discussion of its public health aspects. The incentive for combatting this disease has too often come from the health officer. Many dairymen have gathered the impression that the general acceptance of the view that bovine tuberculosis is not a menace to public health would end their difficulties with tuberculosis. Tuberculosis shortens the life of cows and unquestionably reduces the efficiency of a herd as a machine for producing milk from feed. When its existence may be detected the market value of the animal is obliterated, and the reputation of the breeder of fine stock is injured by general knowledge of its existence in his herd."

The movement to exterminate bovine tuberculosis has already been started in several states, and if California is to do likewise this movement must be undertaken with a full appreciation of the magnitude of the problem. The dairy interests must be educated to the realization of the economic importance of this disease. Dairymen must be shown that it is unprofitable to harbor tuberculous cows in the herd. The nature, cause, and methods of eradication must be fully explained by competent veterinary education. The University of California, through the efforts of its veterinarians, Drs. A. R. Ward and C. M. Haring, as well as this department, have accomplished a little in this line through our limited facilities. However, a systematic plan of instruction should be carried out.

The improvement of our milk supply has not only for its object the eradication of tuberculosis, but also the subject of cleanliness in

dairies. While the sanitation of dairies (eliminating the diseases of cattle) does not come under the supervision of the State Veterinarian, I can not forbear discussing it in this report. This branch of sanitary work is under the control of the State Dairy Bureau. Unfortunately, the bureau is not supplied with sufficient resources to properly control this very important work. Many of our dairies are maintained in a filthy condition and the products from them are inimical to public health. The all important matter of sufficient light, ventilation, and drainage in some of the dairies is utterly neglected. The matter of corral milking can not be too severely condemned.

The subject of dairy sanitation, in its relation to tuberculosis, cannot be overlooked. In order to cope with this disease, and finally exterminate it, it is essential that the environment of the cows should be made healthful. A ground work of sanitary dairies is the first essential, in conjunction with education, in the work of exterminating bovine tuberculosis. This can only be accomplished by adequate legislation and constant official supervision of the dairies.

INSPECTIONS.

During the past two years 17,512 cattle, comprising 315 separate inspections, were made from points in the provisionally quarantined area, and in the quarantined area during the open season, to points outside of the quarantined area in this State.

CONCLUSION.

Since the creation of the offices of State and Assistant State Veterinarian by the legislature of 1899 the demands on our services have constantly increased until at the present day we are unable to meet these demands. We find it necessary with present facilities to devote our energies to the economic problems of cattle tick and sheep scab eradication. This leaves us little time to take care of other very important infectious diseases of livestock. By scattering our efforts little would be accomplished in any one line.

The results of the work in cattle tick and sheep scab eradication show what can be accomplished in other lines if our facilities were adequate. For this reason I would respectfully recommend that this department be supplied with sufficient help to properly cope with the work on hand. The services of one more veterinarian and two field inspectors could be used to great advantage.

Respectfully submitted.

CHARLES KEANE,
State Veterinarian.

REGULATIONS ISSUED DURING THE FIFTY-NINTH FISCAL YEAR PROVIDING FOR THE EXTERMINATION OF SCABIES IN SHEEP.**PROCLAMATION.**

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.
SACRAMENTO, CAL., March 2, 1908.

WHEREAS, The State Veterinarian of the State of California has ascertained that a contagious, communicable disease known as scabies exists generally among the sheep in the State of California, in consequence of which disease the sheep industry of said State is now suffering, and will continue to suffer, serious financial losses unless said disease of scabies in sheep is immediately checked; and

WHEREAS, In consequence of the large number of sheep in said State which are infected with said disease of scabies the secretary of the United States Department of Agriculture has placed in quarantine the entire State of California; and

WHEREAS, Under and by virtue of an Act of the Legislature of the State of California entitled "An Act to protect domestic livestock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this Act, and to provide and appropriation therefor," which became a law March 18, 1899 (Statutes and Amendments to the Codes of the State of California 1899, Chapter XCVI), and as amended March 20, 1905, and March 23, 1907 (Statutes and Amendments to the Codes of the State of California 1905, Chapter CCCLV, and Statutes and Amendments to the Codes of the State of California 1907, Chapter 506), the State Veterinarian of the State of California in order to eradicate and to prevent the further spreading or communication of said disease of scabies in sheep has, on this second day of March, 1908, made and established the following rules and regulations:

(a) All sheep in the State of California must be dipped under the supervision of the State Veterinarian, or his duly authorized agent, between March 15, 1908, and July 15, 1908, inclusive.

(b) Sheep affected with scabies, and other sheep exposed to said disease, must be dipped twice, with an interval between the dippings of not less than nine days nor more than twelve days. All other sheep in the State of California not so affected with scabies, or exposed thereto, must be dipped once.

(c) Dipping solutions must receive the sanction of the said State Veterinarian, or his duly authorized agent, before they will be officially recognized by said officers.

(d) All corrals, chutes, barns, feed lots, premises and other places in which sheep affected with said disease of scabies have been placed or located must be cleaned and disinfected in a manner prescribed by the said State Veterinarian, or his duly authorized agent; and

WHEREAS, The said State Veterinarian, by and with the approval of the undersigned as Governor of the State of California, has established the above rules, orders and regulations for the purposes aforesaid, it is hereby ordered that a violation of any or either of the above rules, orders and regulations shall be an offense and punishable as provided by the laws of the State of California;

Now, therefore, I, J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the above rules, orders and regulations prescribed by said State Veterinarian for the eradication and to prevent the spreading or communication of said disease of scabies in sheep to be in full force and effect, and binding rules, orders and regulations within the State of California, and that a violation thereof shall subject any and all persons so violating any or either of said rules, orders and regulations to the penalties provided

for in section eight of that said Act of the Legislature of the State of California entitled "An Act to protect domestic livestock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this Act, and to provide an appropriation therefor."

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this second day of March, A. D. 1908.

J. N. GILLETT.

Governor of the State of California.

By WARREN R. PORTER,

Lieutenant-Governor and Acting Governor
of the State of California.

[SEAL.]

Attest:

C. F. CURRY,

Secretary of State.

SOUTHERN CATTLE FEVER QUARANTINE REGULATIONS ISSUED DURING THE FIFTY-EIGHTH AND FIFTY-NINTH FISCAL YEARS.

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.

SACRAMENTO, July 15, 1906.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in that certain area of territory situated south and west of the quarantine line hereinafter described are liable to communicate an infectious and contagious disease, known as Texas, splenetic, or southern cattle fever, to cattle located north and east of said line, should said cattle located south and west be shipped, moved or transported, driven or grazed over the lands situated north and east of said line; and

WHEREAS, Under and by virtue of an Act of the Legislature of the State of California, entitled "An Act to amend section two of that certain act of the Legislature of the State of California, entitled 'An Act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this Act, and to provide an appropriation therefor,'" which became a law March 18, 1899, approved March 20, 1905, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenetic or southern cattle fever in cattle, has, on this fifteenth day of July, 1906, made a quarantine line as follows:

Beginning on the Pacific coast where the northern boundary line of San Luis Obispo County connects with the Pacific Ocean; thence easterly along the northern boundary line of San Luis Obispo County to its junction with the western boundary of Kings County; thence northwesterly along the western boundary of Kings and Fresno counties to the northwestern corner of Fresno County; thence northeasterly along the southern boundary of Merced County, and the western, southern and eastern boundaries of those portions of the Chowchilla and Bliss ranches at present included in Madera County, and the southeastern corner of Merced County; thence continuing northeasterly along the northern boundary of Madera County to the northeast corner thereof; thence southerly and easterly along the eastern boundary lines of Madera, Fresno, and Tulare counties to the southeast corner of Tulare County; thence easterly along the southern boundary line of Inyo County to its intersection with the eastern boundary line of the State of California.

WHEREAS, The State Veterinarian has, on this fifteenth day of July, 1906, made and established the following rules and regulations as to the crossing of cattle over the said quarantine line:

From this fifteenth day of July, 1906, to the thirty-first day of October, 1906, inclusive, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting, or from causing or permitting to be driven, led, moved, transported, carried or drifted over or across said quarantine line so established, any cattle originating or being south or west of the said quarantine line, and all railroad and steamship companies, and other transportation companies, their officers, agents and servants, are hereby prohibited from shipping or transporting or accepting for shipment or transportation, any cattle originating or being south or west of said quarantine line, or from shipping or transporting into the State of California any cattle originating or being in that area of territory in the United States south of the Federal quarantine line which has been or may be established by the United States Department of Agriculture, or any cattle infested with the *Boophilus annulatus* tick, except in accordance with the regulations of the United States Department of Agriculture, or except where said cattle are shipped and transported on railroad trains for immediate slaughter, and when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination separate pens shall be set apart to receive them, and no other cattle shall be admitted to said pens. The cars that have carried said stock shall be cleansed and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying cattle from said quarantine area shall bear placards stating that said cars contain "Southern cattle," and each of the waybills of said shipment shall have a note on its face with a similar statement. Whenever any cattle have come from said quarantine area, and shall be reshipped from any point at which they have been reloaded to any other point of destination, the cars carrying said animals shall bear similar placards with like statements, and the waybills so stamped. At whatever point these cattle shall be unloaded they shall be placed in separate pens to which no other cattle shall be admitted.

The cars used to transport such animals, and the pens in which they are fed and watered, and the pens set apart for their reception at points of destination, shall be disinfected and cleaned in the following manner:

(a) Remove all litter and manure; this litter and manure may be disinfected by mixing it with lime or saturating it with a five (5) per cent solution of 100 per cent carbolic acid, or, if not disinfected, it must be stored where no cattle can come in contact with it during the period from February 1, 1906, to October 31, 1906, inclusive.

(b) Wash the cars and feeding and watering troughs with water until clean.

(c) Saturate the entire interior surface of the cars, including the inner surface of the car doors, and the fencing, troughs, chutes, and floors of the pens with a mixture made of one and a half pounds of lime and one-quarter pound of 100 per cent carbolic acid to each gallon of water, or with a solution made by dissolving four (4) ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the above rules and regulations shall be an offense and punishable as provided by the laws of the State of California;

Now, therefore, I, George C. Pardee, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the above boundary of such quarantine within the State of California, and further hereby proclaim the above rules, orders, and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within said state, and I do further proclaim that said rules, orders, and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules, orders or regulations to the penalties provided for in section eight of the said act of the Legislature entitled "An Act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to

carry into effect the provisions of this act, and to provide an appropriation therefor."

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this sixteenth day of July, A. D. 1906.

(Signed) GEO. C. PARDEE,

Governor of the State of California.

Attest:

C. F. CURRY,

Secretary of State.

[SEAL.]

AMENDMENT TO PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.

SACRAMENTO, July 15, 1906.

In consideration of the action of the Boards of Supervisors of the counties of Kern, Tulare, Kings, San Luis Obispo, Fresno, and Madera, in appointing a "County Livestock Inspector," and placing under strict quarantine all areas in said counties known to be infested with the *Boophilus annulatus* tick, the transmitter of Texas, splenetic or southern cattle fever, cattle originating in the above mentioned counties can be moved to points in the State of California north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said animals to be free from contagious and infectious diseases.

It is hereby ordered that, during the continuance of the quarantine line as set forth in the proclamation of July 15, 1906, no cattle originating in the quarantine area as described in said proclamation shall be moved or allowed to move into the counties of Kern, Tulare, Kings, San Luis Obispo, Fresno and Madera.

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be affixed this sixteenth day of July, 1906.

(Signed) GEO. C. PARDEE,

Governor of the State of California.

Attest:

C. F. CURRY,

Secretary of State.

[SEAL.]

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.

SACRAMENTO, CAL., February 1, 1907.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in that certain area of territory situated south and west of the quarantine line hereinafter described are liable to communicate an infectious disease known as Texas, splenetic or southern cattle fever, to cattle located north and east of said line, should said cattle located south and west be shipped, moved or transported, driven or grazed over the lands situated north and east of said line; and

WHEREAS, Under and by virtue of an Act of the Legislature of the State of California, entitled "An Act to amend section two of that certain act of the Legislature of the State of California, entitled 'An Act to protect domestic livestock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor,'" which became a law March 18, 1899, approved March 20, 1905, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenetic or southern cattle fever in cattle, has, on this first day of February, 1907, made a quarantine line as follows:

Beginning on the Pacific coast where the northern boundary line of San Luis Obispo County connects with the Pacific Ocean; thence easterly along the northern boundary line of San Luis Obispo County to its junction with the western boundary of Kings County; thence northwesterly along the western boundary of Kings and Fresno counties to the northwestern corner of Fresno County; thence northeasterly

along the southern boundary of Merced County, and the western, southern and eastern boundaries of those portions of the Chowchilla and Bliss ranches at present included in Madera County; thence continuing easterly along the southern boundary of Merced County to the right of way of the main line of the Southern Pacific Railroad Company; thence southeasterly along the right of way of the main line of the Southern Pacific Railroad Company to the northwestern corner of Tulare County; thence northerly and easterly along the northern boundary line of Tulare County to the northeast corner thereof; thence southerly and easterly along the eastern boundary line of Tulare County to the southeast corner thereof; thence easterly along the southern boundary line of Inyo County to its intersection with the eastern boundary line of the State of California.

WHEREAS, The State Veterinarian has, on the first day of February, 1907, made and established the following rules and regulations as to the crossing of cattle over said quarantine line:

From this first day of February, 1907, to the thirty-first day of October, 1907, inclusive, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting, or from causing or permitting to be driven, led, moved, transported, carried or drifted over or across said quarantine line so established, any cattle originating or being south or west of the said quarantine line, and all railroad and steamship companies, and other transportation companies, their officers, agents and servants, are hereby prohibited from shipping or transporting or accepting for shipment or transportation, any cattle originating or being south or west of said quarantine line, or from shipping or transporting into the State of California any cattle originating or being in that area of territory in the United States south of the Federal quarantine line which has been or may be established by the United States Department of Agriculture, or any cattle infested with the *Boophilus annulatus* tick, except in accordance with the regulations of the United States Department of Agriculture, or except where said cattle are shipped and transported on railroad trains for immediate slaughter, and when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them, and no other cattle shall be admitted to said pens. The cars that have carried said stock shall be cleansed and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying cattle from said quarantine area shall bear placards stating that said cars contain "Southern cattle," and each of the waybills of said shipments shall have a note on its face with a similar statement. Whenever any cattle have come from said quarantine area, and shall be reshipped from any point at which they have been reloaded to any other point of destination, the cars carrying said animals shall bear similar placards with like statements, and the waybills so stamped. At whatever point these cattle shall be unloaded they shall be placed in separate pens to which no other cattle shall be admitted.

The cars used to transport such animals, and the pens in which they are fed and watered, and the pens set apart for their reception at point of destination, shall be cleansed and disinfected in the following manner:

(a) Remove all litter and manure; this litter and manure may be disinfected by mixing it with lime or saturating it with a five (5) per cent solution of 100 per cent carbolic acid, or, if not disinfected it must be stored where no cattle can come in contact with it during the period from February 1, 1907, to October 31, 1907, inclusive.

(b) Wash the cars and feeding and watering troughs with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, and the fencing, troughs, chutes, and floors of the pens with a mixture made of one and a half pounds of lime and one-quarter pound of 100 per cent carbolic acid to each gallon of water; or with a solution made by dissolving four (4) ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the above rules and regulations shall be an offense and punishable as provided by the laws of the State of California;

Now, therefore, I, J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the above boundary of such quarantine within the State of California, and further hereby proclaim the above rules, orders, and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within said state, and I do further proclaim that said rules, orders, and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules, orders, or regulations to the penalties provided for in Section eight of the said act of the Legislature entitled "An Act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this first day of February, A. D. 1907.

J. N. GILLETT.

Governor of the State of California.

Attest:

C. F. CURRY,

Secretary of State.

[SEAL.]

AMENDMENT TO PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.

SACRAMENTO, February 1, 1907.

In consideration of the action of the Boards of Supervisors of the counties of Kern, Tulare, Kings, San Luis Obispo, and Santa Barbara, in appointing a County Livestock Inspector, and placing under strict quarantine all areas in said counties known to be infested with the *Boophilus annulatus* tick, the transmitter of Texas, splenic or southern cattle fever, cattle originating in the above mentioned counties can be moved to points in the State of California north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said cattle to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

In consideration of the action of the Boards of Supervisors of the counties of Madera and Fresno, in appointing a County Livestock Inspector, and placing under strict quarantine all areas in said counties situated west of the right of way of the main line of the Southern Pacific Railroad Company known to be infested with the *Boophilus annulatus* tick, cattle originating in those portions of the above mentioned counties can be moved to points in the State of California north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said animals to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

It is hereby ordered that, during the continuance of the quarantine line as set forth in the proclamation of February 1, 1907, no cattle originating in the quarantine area as described in said proclamation shall be moved or allowed to move into the counties of Kern, Tulare, Kings, San Luis Obispo, Santa Barbara, Fresno and Madera.

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be affixed this fifteenth day of April, A. D. 1907.

J. N. GILLETT.

Governor of the State of California.

Attest:

C. F. CURRY,

Secretary of State.

[SEAL.]

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.
SACRAMENTO, CAL., August 1, 1907.

WHEREAS, The State Veterinarian of the State of California has ascertained that cattle located in that certain area of territory situated south and west of the quarantine line hereinafter described are liable to communicate an infectious disease known as Texas, splenetic or southern cattle fever, to cattle located north and east of said line, should said cattle located south and west be shipped, moved or transported, driven or grazed over the lands situated north and east of said line; and

WHEREAS, Under and by virtue of an Act of the Legislature of the State of California, entitled "An Act to protect domestic live stock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899, amended March 20, 1905, approved March 23, 1907, the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenetic or southern cattle fever in cattle, has, on this first day of August, 1907, made a quarantine line as follows:

Beginning on the Pacific coast where the northern boundary of San Luis Obispo County connects with the Pacific Ocean; thence easterly along the northern boundary of San Luis Obispo County to the northwestern corner of Kern County; thence southeasterly, easterly, and northerly following the western, southern, and eastern boundaries of Kern County to the northeast corner of said county; thence easterly along the northern boundary of San Bernardino County to the one hundred and sixteenth meridian west longitude; thence south following the one hundred and sixteenth meridian west longitude to the southern boundary of San Bernardino County; thence easterly along the southern boundary of San Bernardino County to the eastern boundary of the State of California.

WHEREAS, The State Veterinarian has, on the first day of August, 1907, made and established the following rules and regulations as to the crossing of cattle over said quarantine line:

From this first day of August, 1907, to the thirty-first day of October, 1907, inclusive, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying or transporting, or from causing or permitting to be driven, led, moved, transported, carried or drifted over or across said quarantine line so established, any cattle originating or being south or west of the said quarantine line, and all railroad and steamship companies, and other transportation companies, their officers, agents and servants, are hereby prohibited from shipping or transporting or accepting for shipment or transportation, any cattle originating or being south or west of said quarantine line, or from shipping or transporting into the State of California any cattle originating or being in that area or territory in the United States south of the Federal quarantine line which has been or may be established by the United States Department of Agriculture, or any cattle infested with the *Boophilus annulatus* tick, except in accordance with the regulations of the United States Department of Agriculture, or except where said cattle are shipped and transported on railroad trains for immediate slaughter, and when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them, and no other cattle shall be admitted to said pens. The cars that have carried said stock shall be cleansed and disinfected before they are again used to transport, store or shelter animals or merchandise.

All cars carrying cattle from said quarantine area shall bear placards stating that said cars contain "Southern cattle," and each of the waybills of said shipments shall have a note on its face with a similar statement. Whenever any cattle

have come from said quarantine area, and shall be reshipped from any point at which they have been reloaded to any other point of destination, the cars carrying said animals shall bear similar placards with like statements, and the waybills so stamped. At whatever point these cattle shall be unloaded they shall be placed in separate pens to which no other cattle shall be admitted.

The cars used to transport such animals, and the pens in which they are fed and watered, and the pens set apart for their reception at point of destination, shall be cleansed and disinfected in the following manner:

(a) Remove all litter and manure; this litter and manure may be disinfected by mixing it with lime or saturating it with a five (5) per cent solution of 100 per cent carbolic acid, or, if not disinfected it must be stored where no cattle can come in contact with it during the period from August 1, 1907, to June 1, 1908, inclusive.

(b) Wash the cars and feeding and watering troughs with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, and the fencing, troughs, chutes, and floors of the pens with a mixture made of one and a half pounds of lime and one-quarter pound of 100 per cent carbolic acid to each gallon of water; or with a solution made by dissolving four (4) ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the above rules and regulations shall be an offense and punishable as provided by the laws of the State of California:

Now, therefore, I, J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the above boundary of such quarantine within the State of California, and further hereby proclaim the above rules, orders, and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within said state, and I do further proclaim that said rules, orders, and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules, orders, or regulations to the penalties provided for in section eight of the said act of the Legislature, entitled "An Act to protect domestic livestock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this first day of August, A. D. 1907.

J. N. GILLETT.

Governor of the State of California.

Attest:

C. F. CURRY.

Secretary of State.

[SEAL.]

AMENDMENT TO PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.

SACRAMENTO, August 1, 1907.

In consideration of the action of the Boards of Supervisors of the counties of San Luis Obispo, Santa Barbara and Los Angeles, in appointing a County Livestock Inspector, and placing under strict quarantine all areas in said counties known to be infested with the *Boophilus annulatus* tick, the transmitter of Texas, splenic or southern cattle fever, cattle originating in the above mentioned counties can be moved to points in the State of California north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said animals to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

In consideration of the action of the Boards of Supervisors of the counties of Madera and Fresno, in appointing a County Livestock Inspector, and placing under a strict quarantine all areas in said counties situated west of the right of way of

the main line of the Southern Pacific Railroad Company known to be infested with the *Boophilus annulatus* tick, cattle originating in those portions of the above mentioned counties can be moved to points in the State of California north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said animals to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

In consideration of the action of the Board of Supervisors of the county of San Bernardino, in appointing a County Livestock Inspector, and placing under strict quarantine all areas in said county situated west of the one hundred and sixteenth meridian west longitude, known to be infested with the *Boophilus annulatus* tick, the transmitter of Texas, splenetic or southern cattle fever, cattle originating in the above mentioned part of San Bernardino County may be moved to points in the State of California, north and above the said quarantine line, provided they are first inspected by the State Veterinarian or his duly authorized agent, and are accompanied by a certificate showing said animals to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

It is hereby ordered that, during the continuance of the quarantine line as set forth in the proclamation of August 1, 1907, no cattle originating in the quarantine area as described in said proclamation shall be moved or allowed to move into the counties of San Luis Obispo, Santa Barbara, Los Angeles, Fresno, Madera and San Bernardino.

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be affixed this first day of August, A. D. 1907.

J. N. GILLETT,

Governor of the State of California.

Attest:

C. F. CURRY,

Secretary of State.

[SEAL.]

PROCLAMATION.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.
SACRAMENTO, CAL., February 1, 1908.

WHEREAS. The State Veterinarian of the State of California has ascertained that cattle located in that certain area of territory situated south and west of the quarantine line hereinafter described are liable to communicate an infectious disease known as Texas, splenetic, or southern cattle fever, to cattle located north and east of said line, should said cattle located south and west be shipped, moved or transported, driven or grazed over the lands situated north and east of said line; and.

WHEREAS, Under and by virtue of an Act of the Legislature of the State of California, entitled "An Act to protect domestic livestock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor," which became a law March 18, 1899 (Statutes and Amendments to the Codes of the State of California, 1899, Chapter XCVI), and as amended March 23, 1907 (Statutes and Amendments to the Codes of the State of California, 1907, Chapter 506), the State Veterinarian of the State of California, in order to prevent the spreading or communication of said disease of Texas, splenetic, or southern cattle fever in cattle, has, on this first day of February, 1908, made a quarantine line as follows:

Beginning on the Pacific coast where the northern boundary of San Luis Obispo County connects with the Pacific Ocean; thence easterly along the northern boundary of San Luis Obispo County to the northwestern corner of Kern County; thence southeasterly and easterly, and northerly following the western, southern, and eastern boundaries of Kern County to the northeast corner of said county; thence

easterly along the northern boundary of San Bernardino County to the one hundred and sixteenth meridian west longitude; thence south following the one hundred and sixteenth meridian west longitude to the southern border of San Bernardino County; thence easterly along the southern boundary of San Bernardino County to the eastern boundary of the State of California.

WHEREAS. The State Veterinarian has, on this first day of February, 1908, made and established the following rules and regulations as to the crossing of cattle over said quarantine line:

From this first day of February, 1908, to the thirty-first day of October, 1908, inclusive, every person, company, corporation, their agents and servants, are hereby prohibited from driving, leading, moving, carrying, or transporting, or from causing or permitting to be driven, led, moved, transported, carried, or drifted over or across said quarantine line so established, any cattle originating or being south or west of the said quarantine line, and all railroad and steamship companies, and other transportation companies, their officers, agents and servants, are hereby prohibited from shipping or transporting, or accepting for shipment or transportation, any cattle originating or being south or west of said quarantine line, or from shipping or transporting into the State of California any cattle originating or being in that area of territory in the United States south of the Federal quarantine line which has been or may be established by the United States Department of Agriculture, or any cattle infested with the *Boophilus annulatus* tick, except in accordance with the regulations of the United States Department of Agriculture, or except where said cattle are shipped and transported on railroad trains for immediate slaughter, and when so shipped or transported by railroad the following regulations must be observed:

On unloading said cattle at their destination, or for food and rest en route to same, separate pens must be set apart to receive them, and no other cattle shall be admitted to said pens. The cars that have carried said stock shall be cleansed and disinfected before they are again used to transport, store, or shelter animals or merchandise.

All cars carrying cattle from said quarantine area shall bear placards stating that said cars contain "Southern cattle," and each of the waybills of said shipments shall have a note on its face with a similar statement. Whenever any cattle have come from said quarantine area, and shall be reshipped from any point at which they have been reloaded to any other point of destination, the cars carrying said animals shall bear similar placards with like statements, and the waybills so stamped. At whatever point these cattle shall be unloaded they shall be placed in separate pens to which no other cattle shall be admitted.

The cars used to transport such animals, and the pens in which they are fed and watered, and the pens set apart for their reception at point of destination, shall be cleaned and disinfected in the following manner:

(a) Remove all litter and manure; this litter and manure may be disinfected by mixing it with lime or saturating it with a five (5) per cent solution of 100 per cent carbolic acid; or, if not disinfected, it must be stored where no cattle can come in contact with it during the period from February 1, 1908, to October 31, 1908, inclusive.

(b) Wash the cars and feeding and watering troughs with water until clean.

(c) Saturate the entire inner surface of the cars, including the inner surface of the car doors, and the fencing, troughs, chutes, and floors of the pens with a mixture made of one and a half gallons of lime and one-quarter pound of 100 per cent carbolic acid to each gallon of water; or with a solution made by dissolving four (4) ounces of chloride of lime to each gallon of water.

It is further ordered that a violation of any or either of the above rules and regulations shall be an offense, and punishable as provided by the laws of the State of California.

Now, therefore, I, J. N. Gillett, as Governor of the State of California, by virtue of the authority vested in me by law, do hereby proclaim the above boundary

of such quarantine within the State of California, and further hereby proclaim the above rules, orders, and regulations prescribed by the State Veterinarian for the maintenance and enforcement of such quarantine to be legal and binding rules and regulations within said state, and I do further proclaim that said rules, orders, and regulations shall be maintained and enforced within the State of California, and that a violation thereof shall subject all persons so violating any of said rules, orders, or regulations to the penalties provided for in section eight of that said act of the legislature entitled "An act to protect domestic livestock from contagious and infectious diseases, to provide for the appointment and duties of officials to carry into effect the provisions of this act, and to provide an appropriation therefor."

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this first day of February, A. D. 1908.

(Signed) J. N. GILLETT,

Governor of the State of California.

Attest: _____
(Signed) C. F. CURRY,
[SEAL.] Secretary of State.

AMENDMENT No. 1 TO PROCLAMATION OF FEBRUARY 1, 1908.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.
SACRAMENTO, February 1, 1908.

In consideration of the action of the Boards of Supervisors of San Luis Obispo, Santa Barbara, Ventura, Los Angeles, and Riverside counties, in appointing County Livestock Inspectors, and placing under strict quarantine all areas in said counties known to be infested with the *Boophilus annulatus* tick, the transmitter of Texas, splenetic, or southern cattle fever, cattle originating in the above mentioned counties may be moved to points in the State of California north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said cattle to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

In consideration of the action of the Board of Supervisors of San Bernardino County in appointing a County Livestock Inspector, and placing under strict quarantine all areas in said county situated west of the one hundred and sixteenth meridian west longitude, known to be infested with the *Boophilus annulatus* tick, the transmitter of Texas, splenetic or southern cattle fever, cattle originating in the above mentioned part of San Bernardino County may be moved to points in the State of California, north and above the said quarantine line, provided they are first inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate showing said animals to be free from contagious and infectious diseases and the *Boophilus annulatus* tick.

It is hereby ordered, that during the continuance of the quarantine line as set forth in the proclamation of February 1, 1908, no cattle originating in the quarantine area as described in said proclamation shall be moved or allowed to move into the counties of San Luis Obispo, Santa Barbara, Ventura, Los Angeles, Riverside, and San Bernardino.

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be affixed this first day of February, A. D. 1908.

(Signed) J. N. GILLETT,

Governor of the State of California.

Attest: _____
(Signed) C. F. CURRY,
[SEAL.] Secretary of State.

AMENDMENT No. 2 TO PROCLAMATION OF FEBRUARY 1, 1908.

EXECUTIVE DEPARTMENT, STATE OF CALIFORNIA.

WHEREAS, The State Veterinarian of the State of California has ascertained that certain cattle located in that part of Fresno County situated west of the main line of the Southern Pacific Railroad Company are infested with *Boophilus annulatus* tick, the transmitter of Texas, splenetic, or southern cattle fever, and are liable to communicate this disease of Texas, splenetic, or southern cattle fever to cattle located in the State of California outside of this part of Fresno County, should said cattle located in that part of Fresno County situated west of the main line of the Southern Pacific Railroad Company be shipped, moved, or transported, driven, or grazed over the lands in the State of California situated outside of said part of Fresno County; therefore,

It is hereby ordered, that cattle located in that part of Fresno County situated west of the main line of the Southern Pacific Railroad must not be moved or allowed to move outside of said part of Fresno County to any part of the State of California, unless said cattle have first been inspected by the State Veterinarian, or his duly authorized agent, and are accompanied by a certificate issued by such inspector stating that said cattle are free from contagious and infectious diseases, and the *Boophilus annulatus* tick.

It is further ordered that a violation of any or either of the above rules and regulations shall be an offense, and punishable as provided by the laws of the State of California, as set forth in the proclamation of February 1, 1908.

In Witness Whereof, I have hereunto set my hand and caused the Great Seal of this State to be hereunto affixed, this first day of February, A. D. 1908.

(Signed) J. N. GILLETT,
Governor of the State of California.

Attest:
(Signed) C. F. CURRY,
[SEAL.] Secretary of State.